

Quiz 3 (30 minutes)

Name:

ID:

Problem 2 (30 points) Find the Taylor expansion of the following at $z = 1$

$$f(z) = 1 + z^2 + \sin(\pi z)$$

$$\begin{aligned}
 f(z) &= 1 + z^2 + \sin \pi z \\
 &= (z-1)^2 + 2(z-1) + 2 + \sin \pi z \\
 &= (z-1)^2 + 2(z-1) + 2 - \sin(\pi(z-1)) \\
 &= (z-1)^2 + 2(z-1) + 2 - \left(\pi(z-1) - \frac{\pi(z-1)^3}{3!} + \frac{\pi(z-1)^5}{5!} \dots \right) \\
 &= (z-1)^2 + 2(z-1) + 2 - \sum_{n=0}^{\infty} \frac{(-1)^n [\pi(z-1)]^{2n+1}}{(2n+1)!} \\
 &= (z-1)^2 + 2(z-1) + 2 + \sum_{n=0}^{\infty} \frac{(-1)^{n+1} [\pi(z-1)]^{2n+1}}{(2n+1)!}
 \end{aligned}$$

Problem 1 (40 points): Find the Laurent expansion of

$$(1) f(z) = \frac{z}{(z^2-4)^2} \text{ where } 0 < |z-2| < 4.$$

$$(2) f(z) = \frac{3z}{(z-1)^2(z-2)}$$

(1)

$$\begin{aligned}
 f(z) &= \frac{z}{(z^2-4)^2} \\
 &= \frac{1}{8} \left(\frac{1}{(z-2)^2} - \frac{1}{(z+2)^2} \right) \\
 &= \frac{1}{8} \frac{1}{(z-2)^2} - \frac{1}{8} \left(\frac{1}{4 \left(1 - \frac{-(z-2)}{4} \right)} \right)^2 \\
 &= \frac{1}{8} \frac{1}{(z-2)^2} - \frac{1}{128} \left(\sum_{n=0}^{\infty} (-1)^n \left(\frac{z-2}{4} \right)^n \right)^2
 \end{aligned}$$

$$\begin{aligned}
 (2) f(z) &= \frac{3z}{(z-1)^2(z-2)} = \frac{A}{z-2} + \frac{B}{z-1} + \frac{C}{(z-1)^2} \\
 &= -\frac{3}{(1-z)^2} + \frac{6}{1-z} - \frac{6}{2-z} \\
 |z| < 1 \quad f(z) &= -3 \left(\sum_{n=0}^{\infty} z^n \right)^2 + 6 \sum_{n=0}^{\infty} z^n - 3 \sum_{n=0}^{\infty} \left(\frac{z}{2} \right)^n \\
 1 < |z| < 2 \quad f(z) &= -3z^{-2} \left(\sum_{n=0}^{\infty} \left(\frac{1}{z} \right)^n \right)^2 - 6z^{-1} \sum_{n=0}^{\infty} \left(\frac{1}{z} \right)^n - 3 \sum_{n=0}^{\infty} \left(\frac{z}{2} \right)^n \\
 |z| > 2 \quad f(z) &= -3z^{-2} \left(\sum_{n=0}^{\infty} \left(\frac{1}{z} \right)^n \right)^2 - 6z^{-1} \sum_{n=0}^{\infty} \left(\frac{1}{z} \right)^n + 6z^{-1} \sum_{n=0}^{\infty} \left(\frac{2}{z} \right)^n
 \end{aligned}$$

Problem 3 (30 points) Determine the property of the singularity of the following functions. For poles, determine their orders. (判断下面函数孤立奇点的性质, 如果是极点, 确定其阶数)

$$(1) \frac{z^3 + 2z + 1}{(z-3)^2(z-2)}$$

$$(2) \frac{\sin z}{z^3} - \frac{1}{z^2}$$

(1) $z = 3$ 为二阶极点, $z = 2$ 为一阶极点

(2) $z = 0$ 为可去奇点, $z = \infty$ 为本性奇点

(1) $z = 3$ is a second-order pole, $z = 2$ is a first-order pole

(2) $z = 0$ is a removable singularity, $z = \infty$ is an essential singularity

数学物理方法 P77:

本性奇点: 函数在本性奇点邻域内的 Laurent 展开具有无穷多个负幂项

Essential singularity: The Laurent expansion of a function in the neighborhood of an essential singularity has infinite negative power terms.

$$f(z) = \frac{\sin z}{z^3} - \frac{1}{z^2}$$

$$= \frac{\sin z - z}{z^3}$$

$$= \frac{z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots + (-z)}{z^3}$$

$$f(z) = -\frac{1}{3!} + \frac{z^2}{5!} - \frac{z^4}{7!} + \cdots$$

$$t = \frac{1}{z}$$

$$f\left(\frac{1}{t}\right) = -\frac{1}{3!} + \frac{1}{5!}t^{-2} - \frac{1}{7!}t^{-4} + \cdots$$

0 是 $f(1/t)$ 的本性奇点，所以 ∞ 是 $f(z)$ 的本性奇点。

因为 ∞ 点 扣分 会统一加的

END